

An Introduction To EDA TOOLS



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In part 1 of this article we learnt about LiveWire and Multisim electronic design automation (EDA) tools. Let us now take a look at OrCAD and Proteus.

OrCAD

It is one of the most-popular EDA tools in industry. OrCAD is a suite of tools from Cadence for design, simulation and layout of PCBs. This software is mainly used by design engineers to create electronic schematics and electronic prints for manufacturing PCBs. It enables designing of a circuit from beginning to end. It is highly-customisable; we can alter the components and their characteristics based on our requirements.

OrCAD comes with three main applications. Capture, the first application, is used to draw a circuit on the screen, known as schematic capture. It offers great flexibility when compared with traditional paper-and-pen drawing, as design changes can be incorporated without difficulty and error corrections can be done easily.

The second application, PSpice, simulates the captured circuit. We can analyse the circuit's behaviour in different ways and verify that it performs as required.

PCB Editor, the last, is used to design PCBs. Output of this is a set of files that can be sent to the manufacturer

for PCB fabrication or to an electronics workshop. Go through these applications before using any of the tools, as it will improve your learning skills.

The program used here is a version of Simulation Program for Integrated Circuit Engineering (SPICE), which was developed at University of California in Berkeley, USA, in the 1970s. For many years it has been the most-widely-used circuit simulator in the electronics industry. It has many versions, of which we will describe PSpice.

There are three basic steps to be followed:

1. Draw an electronic circuit using Capture.
2. Simulate it with PSpice using specific models for the devices and analyse its behaviour with Probe.
3. Create basic design steps for the PCB using PCB Editor.

Following are the steps used in OrCAD (16.0 version) for simulation and PCB design:

1. Open OrCAD program on the computer. The file will be in the folder of installed software named OrCAD Capture.
2. Opening the tool will direct you to a Cadence product choice window. Select OrCAD Capture and click OK. It will lead you to a new window of Capture.
3. Open a new project from File menu (Fig. 12).
4. Provide the details in the next window and select Analog or Mixed A/D (Fig. 13).
5. Giving project details will lead to a

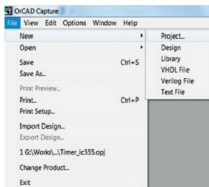


Fig. 12: Creating a new project



Fig. 13: New project window

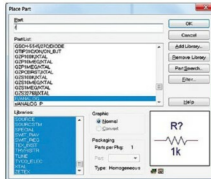


Fig. 14: Placing the components

